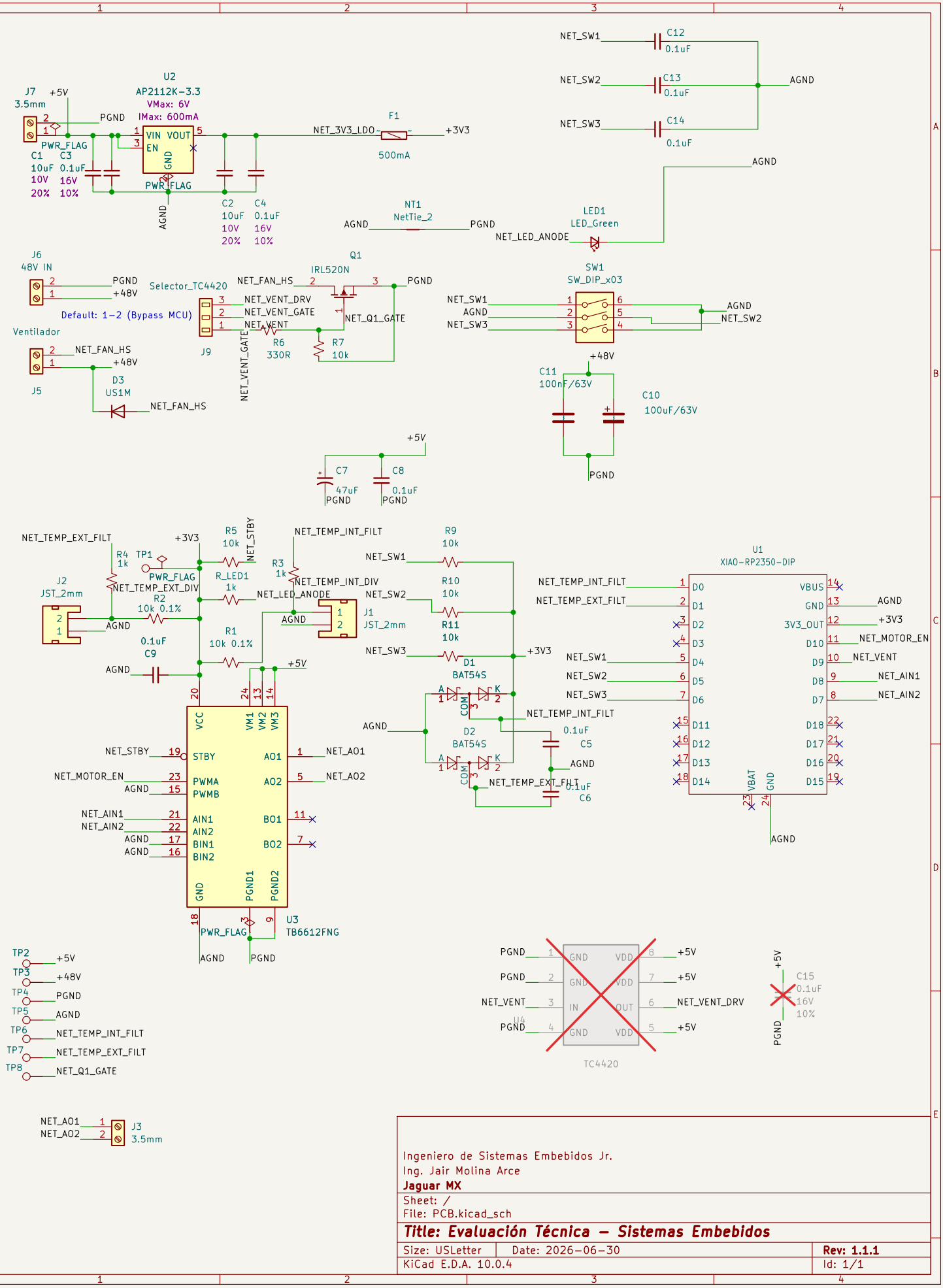
U2
AP2112K-3.3
VMax: 6V
IMax: 600mA
J7
3.5mm
+5V
PGND
PWR_FLAG
C1
10uF
10V
20%
C3
0.1uF
16V
10%
VIN
VOUT
EN
GND
PWR_FLAG
AGND
C2
10uF
10V
20%
C4
0.1uF
16V
10%
F1
NET_3V3_LDO
+3V3
500mA
NET_SW1
NET_SW2
NET_SW3
C12
C13
C14
0.1uF
AGND
NT1
NetTie_2
LED1
LED_Green
NET_LED_ANODE
J6
48V IN
PGND
+48V
Default: 1-2 (Bypass MCU)
Ventilador
J5
D3
US1M
NET_FAN_HS
Selector_TC4420
J9
NET_VENT_DRV
NET_VENT_GATE
NET_VENT
R6
330R
R7
10k
Q1
IRL520N
NET_Q1_GATE
SW1
SW_DIP_x03
C11
100nF/63V
C10
100uF/63V
C7
47uF
C8
0.1uF
J2
JST_2mm
TP1
R4
1k
NET_TEMP_EXT_FILT
NET_TEMP_EXT_DIV
R2
10k 0.1%
R1
10k 0.1%
C9
0.1uF
R5
10k
R_LED1
1k
NET_STBY
R3
1k
NET_TEMP_INT_FILT
NET_TEMP_INT_DIV
J1
JST_2mm
R9
10k
R10
10k
R11
10k
D1
BAT54S
D2
BAT54S
C5
0.1uF
C6
0.1uF
U3
TB6612FNG
VCC
VM1
VM2
VM3
STBY
PWMA
PWMB
AIN1
AIN2
BIN1
BIN2
AO1
AO2
BO1
BO2
GND
PGND1
PGND2
PWR_FLAG
U1
XIAO-RP2350-DIP
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
D16
D17
D18
VBUS
3V3_OUT
VBAT
TP2
TP3
TP4
TP5
TP6
TP7
TP8
U4
TC4420
C15
0.1uF
16V
10%
J3
3.5mm
Ingeniero de Sistemas Embebidos Jr.
Ing. Jair Molina Arce
Jaguar MX
Sheet: /
File: PCB.kicad_sch
Title: Evaluación Técnica – Sistemas Embebidos
Size: USLetter
Date: 2026-06-30
Rev: 1.1.1
KiCad E.D.A. 10.0.4
Id: 1/1